

# Statistical Analysis Plan (SAP)

Swiss multi-centre, randomized, placebo controlled trial of pregabalin for prevention of persistent pain in high risk patients undergoing breast cancer surgery

## PREVENT

### Administrative information

|                            |                            |
|----------------------------|----------------------------|
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### Study information

|                             |                                                                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title of the study          | Swiss multi-centre, randomized, placebo controlled trial of pregabalin for prevention of persistent pain in high risk patients undergoing breast cancer surgery |
| Principal Investigator (PI) | Benno Rehberg-Klug                                                                                                                                              |
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## 1. Timing of SAP writing

### 1.1 Scope

The present SAP cover the primary and secondary analyses of the PREVENT trial, for both the main study and the ancillary study. It does not cover the interim analysis which was conducted prior to the finalization of the SAP.

Although, in the PREVENT study, genetic material will be collected for a later genetic association analysis on acute and chronic post-surgical pain, the analysis of these data is beyond the scope of the present SAP.

### 1.1 Timing

The current version of the SAP was written and finalized **after database lock** and before the conduct of analyses.

## 2. Introduction

### 2.1 Background and rationale

Persistent postsurgical pain occurs in more than 30% of patients undergoing breast cancer surgery. Evidence that gabapentinoids such as pregabalin may reduce the incidence of persistent postsurgical pain is ambiguous, potentially because in previous trials prophylactic treatment was administered to every patient undergoing surgery. The patients at low risk of long-term pain were exposed to side effects without much benefit to expect.

### 2.2 Objectives

Main study

Aim: Testing the efficacy of pregabalin to prevent long term postoperative pain in patients at high risk of persistent pain after breast cancer surgery.

Ancillary study

Aim: Analysing how side-effect information influences treatment tolerance.

## 3. Study methods

### 3.1 Study population

The study will take place in 3 academic hospitals, 1 cantonal hospital and 3 private hospitals, all located in Switzerland:

- University Hospital Bern, Inselspital, 3010 Bern
- Centre Hospitalier Universitaire Vaudois CHUV, 1011 Lausanne
- Hôpitaux Universitaires de Genève HUG, 1211 Genève

- Swiss Medical Network : Clinique de Genolier, 1272 Genolier and Clinique Générale Ste. Anne, 1700 Fribourg
- Clinique des Grangettes, 1224 Chêne-Bougeries (GE)
- Brustzentrum Bern, 3013 Bern
- Kantonsspital St. Gallen, 9007 St.Gallen
- Patients will be recruited from the patient population speaking German, English, or French. All 5 hospitals have large breast-cancer centres, thus providing a large patient base for recruitment. If initial enrolment goals are not met, the inclusion period will be prolonged.

### 3.2 Inclusion/exclusion criteria

Participants fulfilling all of the following inclusion criteria are eligible for the study:

- patients scheduled for breast surgery for cancer, either breast-conserving (tumorectomy/quadrantectomy) or mastectomy, with or without immediate reconstruction, and with or without axillary dissection.
- patients of 18 years or more scheduled for above mentioned type of surgery
- ability to speak and read French, English or German
- high (>30%) risk of clinically important persistent pain: identified with 2 or more points of a risk score including the items: pre-existing pain at surgical site (2 points), history of depression (1 point), age < 50 years (1 point), and high expected acute pain (>6/10, 1 point).
- Informed Consent as documented by signature (Appendix Informed Consent Form)
- The presence of any one of the following exclusion criteria will lead to exclusion of the participant:
  - Inability to understand the consent form and to follow the procedures of the study, e.g. due to language problems, psychological disorders, dementia, etc. of the participant
  - Participation in another study with investigational drug within the 30 days preceding and during the present study,
  - Previous enrolment into the current study,
  - Enrolment of the investigator, his/her family members, employees and other dependent persons
  - Pregnancy or lactation
  - renal insufficiency (creatinine clearance < 60 ml/min using the formula  $CL_{er}(min/min) = (1.23 * (140 - age) * weight / serum\ creatinine\ in\ \mu mol/l) * 0.85$ )
  - Allergy to pregabalin or the ingredients of the capsules
  - Long-term preoperative therapy with gabapentinoids or high-dose opioids (more than 60 mg of morphine equivalents)
  - Symptomatic cardiac insufficiency (peripheral oedema, NYHA class III – marked limitation of physical activity)
  - Suicidal ideation, identified by the question: “have you been bothered by thoughts that you would be better off dead, or of killing yourself?”
  - Planned fertility preservation immediately after surgery before a planned chemotherapy
  - Known or suspected non-compliance, or substance-use disorder with impact on medication adherence

Patients will be screened for the study at the preoperative anaesthesia or gynaecological consultation by the anaesthesiologist or gynaecologist performing the respective consultation.

Patients with a high (>30%) risk of clinically important persistent pain will be identified with 2 or more points of a risk score developed in a previous study 2:

- pre-existing pain at surgical site (2 points),
- history of depression (1 point),
- age < 50 years (1 point),
- high expected acute pain (>6/10, 1 point).

Only patients identified by this risk score as being at high risk will be included in the full randomized placebo-controlled study protocol. To create a control group of patients with a lower risk of persistent pain for the genetic association study (see "other outcomes of interest"), all patients screened for the study will be asked to participate in this observational genetic association study, without participating in the randomized controlled trial. A separate patient information and consent form will be used for this genetic association study. The same separate patient information and consent form will be given to patients participating in the randomized controlled trial. Patients participating in the randomized controlled trial will not be obliged to participate in the genetic association study.

At the consultation, patients will be informed orally and receive written information about the study. If they are a priori willing to accept, a blood test for creatinine clearance and a urine test for pregnancy will be performed (if not already existing or ordered for medical purposes). These patients will be contacted by a study nurse or investigator to arrange a separate appointment for explanation of the study procedures and distribution of the study medication.

### 3.3 Design

Randomized, double-blind, placebo-controlled study

### 3.4 Randomization

The study statistician will supply a computer-generated randomization list for all hospitals. Allocation ratio will be 1:1, and there will be a block randomization in blocks of 4 and 8 patients. The randomization list will be kept concealed by the pharmacy of the HUG until the end of the study (last visit of the last patient) and completion of data entry into the database.

A second computer-generated randomization list will be prepared by the study statistician for the four different versions of the side effect information. It will be stratified by center and generated by blocks of sizes 4, 8, and 12.

### 3.5 Blinding

The pharmacy of the HUG will provide numbered treatment packs with neutral labelling of the study drug, so that all care providers as well as the trial participants will be blinded to the study. Identification of the treatment according to the randomization list will take place only after the end of the study and completion of data entry into the database. Investigators will rest blinded during data analysis. Only the statistician, who is independent of the investigators, will have access to the randomization list.

The attribution to the different side effect information versions will be randomized and the study team interacting with patients will be blinded. The participants will be blind to the fact that there are different information sets, and all patients will receive a first information about potential

side effects before inclusion in the study, with the study patient information. Four different versions of information sheets will be prepared: 1) a version mentioning only the most frequent side effect (Fatigue, dizziness, drowsiness, blurred vision), 2) a version mentioning all subjective side effects, 3) a version mentioning the most frequent side effects and the notion that these occur also in placebo groups, and 4) a version mentioning all subjective side effects and the notion that these occur also in placebo groups.

Patients will receive one of these versions (randomized between patients) and the text will be read out loud by an investigator at the moment the patient receives the study drug.

In case of a serious adverse event with potential relation to the study drug (including hypersensitivity reactions including angioedema), and only in this case, investigators or physicians treating participants may call the pharmacy of the HUG via an emergency number for unblinding of the study drug of this patient. Only the pharmacy of the HUG has access to the randomization list. Further administration of the study drug will be stopped.

### 3.6 Sample size, null and alternative hypothesis

The primary objective of this study is to show a lower risk of clinically important pain at 3 months in patients treated with pregabalin compared to placebo.

**H<sub>0</sub>:** The incidence of clinically important pain at 3 months postoperative is the same in the pregabalin group and the placebo group

**H<sub>1</sub>:** The incidence of clinically important pain at 3 months postoperative is different between the pregabalin group and the placebo group.

The sample size was calculated to detect a difference of 20% in the risk (i.e. incidence), assuming a power of 80% and a two-sided risk alpha of 0.05.

In our previous study, a model for the risk of clinically important pain at 4 months was established, with the risk factors preoperative pain at the surgical site, depression, age <50 years, and high expected pain (>6/10). Using a score of this risk factors, a subpopulation of patients with a predicted risk of 30% or higher of clinically important persistent pain can be identified. This subgroup represented 36.1% of the enrolled patients and the actual incidence of clinically important pain at 4 months was 57.6%.

In the present study, patients will be recruited in this high risk sub-population and we assume that the incidence in the control/placebo group is 57.6% (probably it is somewhat higher, since this was the incidence at 4 months in the previous study, and in the planned study the incidence is measured at 3 months, to be conform to IMMPACT recommendations).

A decrease of the incidence of clinically important pain of at least 20% is desirable for the intervention to be clinically interesting. In the studies summarized in the meta-analysis of Clarke, the pooled odds ratio for persistent pain for pregabalin was 0.09 (0.02-0.79, only 2 trials). Due to the small sample size of existing studies, the risk reduction effect is probably overestimated, but 20% reduction in incidence appears to be realistic.

To detect a difference in the incidence of clinically important pain of 20%, the necessary sample size is 95 per arm (total 190).



An intermediary analysis is planned when half the patients have been recruited to test for futility of continuation. To compensate for the loss of statistical power, 115 patients per arm are needed for the final analysis.

Retention rates in pregabalin long-term-treatment are low, due to CNS-related side effects in particular. Data are available only for the indication as an antiepileptic and in generalized anxiety. These data indicate a retention rate of 75% at 6 months and 50% at 1 year. We assume an attrition rate during treatment of 15-20% in our study. In addition, we expect a post-treatment drop-out rate at 3 months of 10%. Therefore, the necessary actual sample size will be about 150 per arm (total 300). We will continue inclusion until 126 patients per arm have completed the treatment phase of the study.

### 3.7 Interim analyses and stopping rules

An intermediary analysis is planned when half the patients have been recruited to test for futility of continuation. To compensate for the loss of statistical power, 115 patients per arm are needed for the final analysis.

The interim analysis will be performed by the study statistician when data for the primary outcome are available for 60 patients per arm.

Stopping rules are a difference in the incidence of the primary outcome of less than 8.2% (futility) or more than 19.9% (superiority).

### 3.9 Timing of outcome assessment (timepoints)

| Study periods                                                                       | Screening     | Inclusion     | Treatment |         |               | Follow-up      |                 |                   |                   |
|-------------------------------------------------------------------------------------|---------------|---------------|-----------|---------|---------------|----------------|-----------------|-------------------|-------------------|
|                                                                                     | Visit 1       | Visit 2       | Visit 3   | Visit 4 | Visit 5       | Visit 6        | Visit 7         | Visit 8           | visit 9           |
|                                                                                     | Day -30 to -2 | Day -29 to -1 | Day 0     | Day 1   | Day 10 (7-12) | Day 30 (20-35) | Day 90 (90-120) | Day 180 (150-210) | Day 360 (330-390) |
| In/exclusion criteria                                                               | x             |               |           |         |               |                |                 |                   |                   |
| Patient information by surgeon or                                                   | x             |               |           |         |               |                |                 |                   |                   |
| laboratory tests: creatinine clearance, pregnancy                                   | (x)           | x             |           |         |               |                |                 |                   |                   |
| Explanation of effects and side effects of the IMP (randomized neutral or positive) |               | x             |           |         |               |                |                 |                   |                   |
| Informed consent                                                                    |               | x             |           |         |               |                |                 |                   |                   |
| Randomisation of IMP treatment group                                                |               | x             |           |         |               |                |                 |                   |                   |
| preoperative questionnaires                                                         | x             | x             |           |         |               |                |                 |                   |                   |
| Dispensing of study drug                                                            |               | x             |           |         |               |                |                 |                   |                   |
| Study treatment                                                                     |               | (x)           | x         | x       | x             |                |                 |                   |                   |
| recording of perioperative analgesia                                                |               |               | x         |         |               |                |                 |                   |                   |
| recording of postoperative analgesia                                                |               |               |           | x       |               |                |                 |                   |                   |
| Secondary outcome recording:                                                        |               |               |           |         |               |                |                 |                   |                   |
| acute pain: questionnaire pain-out                                                  |               |               |           | x       |               |                |                 |                   |                   |
| treatment adherence                                                                 |               |               | x         | x       | x             | x              |                 |                   |                   |
| pain diary (pain intensity, concomitant treatment)                                  |               |               |           | x       | x             | x              |                 |                   |                   |
| questionnaire BPI                                                                   |               |               |           |         |               |                | x               | x                 | x                 |
| neuropathic pain (questionnaire DN4)                                                |               |               |           |         |               | x              | x               | x                 | x                 |
| expectations about treatment effect                                                 |               | x             |           |         | x             |                |                 |                   |                   |
| belief about treatment attribution                                                  |               |               |           |         | x             |                |                 |                   |                   |
| Concomitant medication                                                              |               |               | x         | x       | x             | x              | x               | x                 | x                 |
| Side effect, adverse event recording                                                |               |               | x         | x       | x             |                |                 |                   |                   |
| Primary outcome recording: questionnaire BPI                                        |               |               |           |         |               |                | x               |                   |                   |

### 3.10 Timing of final analysis

All outcomes will be analyzed after study completion. After completion of data entry, data validation and cleaning will be performed. Data analysis will start after database lock.

### 3.11 Data management

All data are extracted from REDCap and stored on the HUG server to which only the UAM staff have access. When an item exists in several languages (English, German, French), a new variable will be created to reduce the information from all three matching variables into one.

## 4. Analysis population

### 4.1 Full analysis set

The Full Analysis Set follows the intention-to-treat (ITT) principle and will include all randomized participants, analyzed according to the treatment group to which they were randomized, regardless of treatment actually received, protocol deviations, or study completion status

Participants will only be excluded from the FAS if randomization was performed in error (e.g., the participant never met eligibility criteria and received no study treatment) or if informed consent was withdrawn prior to any data collection. These exclusions will be clearly documented in the participant disposition listing and summarized in the CONSORT flow diagram.

The FAS represents the main analysis population for all efficacy endpoints unless otherwise specified.

### 4.2 Per-protocol set

The Per-Protocol set (PP) comprises randomized participants in the Full Analysis Set (FAS) who completed treatment per protocol, and had no major protocol deviations that could materially affect the assessment of the primary endpoint. Participants in the PP set are analyzed according to their randomized treatment assignment.

### 4.3 Safety population (Safety set/As-Treated)

Includes all participants who received at least one dose of study treatment (Pregabalin or placebo). Participants are analyzed according to the treatment actually received, regardless of randomization. Appendix tables will summarize participants who switched treatments, including timing and duration of exposure.

## 5. Protocol compliance and deviations

### 5.1 Definition

Protocol deviations are defined as departures from the approved study protocol, study-specific procedures, or Good Clinical Practice (GCP) requirements that occur after participant enrollment. Deviations will be identified and documented throughout the study.

## 5.2 Classification

Protocol deviations will be classified as either:

- **Major (important) deviations**, which may have an impact on subject safety, data integrity, or study outcomes (e.g., enrollment of ineligible subjects, failure to obtain informed consent, significant dosing errors, missed critical assessments).  
Major deviation will include the following pre-specified criteria: the absence of 150 mg each of the first four visits and the absence of the approved compliance as assessed on the 13<sup>th</sup> day. Additional major deviations criteria might be included retrospectively.
- **Minor deviations**, which are administrative in nature and are not expected to affect subject safety or data integrity.

## 5.3 Identification and handling

Major deviations will be reviewed for potential impact on inclusion in analysis populations (e.g., Per-Protocol Set). Protocol compliance will be summarized descriptively. The following will be presented: Number and percentage of subjects with **≥1 protocol deviation**, overall and by treatment group.

# 6. Statistical analysis

## 6.1 General consideration

All efficacy analyses will be performed on the full analysis set population, unless otherwise specified. All safety analyses will be performed in the safety-evaluable population, unless otherwise specified. Analyses of demographics and other baseline information will be based on full analysis set population. The baseline value of any variable will be defined as the last available data point prior to the first administration of study drug.

Continuous variables will be summarized using descriptive statistics, including the number of missing observations, mean, standard deviation (SD), median (Q1, Q3), and range (Min, Max).

Categorical and binary variables will be summarized using frequency counts and percentages. Percentages will generally be based on the number of non-missing observations, unless otherwise specified. Categories with zero frequency will be displayed in the tables unless otherwise stated.

Scores with more than five categories will be summarized as continuous variables using appropriate descriptive statistics (number of missing values, mean, SD, median, Q1, Q3, minimum, and maximum). Scores with five or fewer categories will be summarized using frequency counts and percentages.

Count data (e.g., number of events, number of visits) will be summarized using descriptive statistics including the number of missing observations, mean, standard deviation (SD), median, Q1, Q3, minimum, and maximum.

A two-sided p-value of less than 0.05 will be considered statistically significant, unless stated otherwise. Confidence intervals will be reported at a level of 95% two-sided.

## 6.2 Primary endpoint

### 6.2.1 Estimand framework

Following the estimand framework introduced in the ICH-E9 addendum (ICH 2020), the attributes of the estimand built around the primary endpoint are defined as follows:

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Population</b>                                | All randomized participants — Intent-to-treat (ITT). Primary analysis includes all randomized participants part of the full analysis set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Variable</b>                                  | Binary indicator of <i>clinically important pain</i> at 3 months after surgery at the site of surgery (breast, axilla, or arm).<br><br>“Clinically important pain” is defined by: taking analgesics for pain at the surgical site OR average resting pain at the surgical site $\geq 3/10$ OR average movement-induced pain at the surgical site $\geq 5/10$ .                                                                                                                                                                                                                                                                              |
| <b>Treatment</b>                                 | Randomized assignment to Pregabalin versus randomized assignment to placebo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Intercurrent events and Handling Strategy</b> | <ul style="list-style-type: none"><li>– Treatment discontinuation / non-adherence: observed 3-month outcomes are used when available (implicit <i>treatment-policy</i> strategy for observed data).</li><li>– Use of rescue medication / concomitant therapy: observed 3-month outcomes are used when available (implicit <i>treatment-policy</i> strategy for observed data).</li><li>– Death before assessment: participants who die before the 3-month assessment do not contribute a primary-endpoint value and are therefore treated as <i>missing</i>., following a hypothetical strategy as defined by Pham et al. (2025).</li></ul> |
| <b>Population-level summary</b>                  | Difference in incidence (absolute risk difference) of clinically important pain between treatment groups at 3 months.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### 6.2.2 Decision table

| <b>Analgesic use</b> | <b>Resting pain</b> | <b>Movement pain</b> | <b>Derived endpoint</b> | <b>Rule explanation</b>               |
|----------------------|---------------------|----------------------|-------------------------|---------------------------------------|
| Missing              | Missing             | Missing              | Missing                 | No information available              |
| Yes                  | Any                 | Any                  | Positive (1)            | Analgesic use alone triggers positive |
| No                   | $\geq 3$            | Any                  | Positive (1)            | Pain threshold met at rest            |
| No                   | Any                 | $\geq 5$             | Positive (1)            | Pain threshold met during movement    |
| No                   | $< 3$               | $< 5$                | Negative (0)            | Pain thresholds not met               |
| No                   | Missing             | $< 5$                | Negative (0)            | Available info below threshold        |

|         |         |         |              |                                               |
|---------|---------|---------|--------------|-----------------------------------------------|
| No      | <3      | Missing | Negative (0) | Available info below threshold                |
| Missing | ≥3      | Any     | Positive (1) | Pain threshold met from available scores      |
| Missing | Any     | ≥5      | Positive (1) | Pain threshold met from available scores      |
| Missing | <3      | <5      | Negative (0) | Available scores below thresholds             |
| Missing | Missing | <5      | Negative (0) | Only movement pain available, below threshold |
| Missing | <3      | Missing | Negative (0) | Only resting pain available, below threshold  |

### 6.2.3 Analysis method

To test the effect of treatment on the risk of clinically important pain 3 months post-surgery while accounting for center, a Mantel–Haenszel (MH) common risk difference test will be used. Clinically important pain at 3 months (Yes/No) will be the outcome, treatment (Pregabalin vs. Placebo) the predictor of interest, and the analysis stratified by center. The pooled risk difference (RD) will be reported with a Wald-type 95% confidence interval and the corresponding p-value.

Prior to interpretation, a test for homogeneity of stratum-specific risk differences will be performed (e.g., weighted least squared method proposed by Lipsitz et al. (1998)).

If the test indicates homogeneous risk differences across strata, the MH risk difference will be interpreted as an estimate of a common treatment effect.

If the test indicates heterogeneous risk differences, the MH risk difference will be interpreted as a weighted average of stratum-specific effects, and stratum-specific risk differences will also be presented.

### 6.2.4 Missing data imputation and sensitivity analysis

For the primary analysis, the full analysis set will be used in the analysis. Missing values will be imputed using the predictive mean matching (PMM) method. This method assumes a missing at random situation. The variables used in the dataset being imputed will be: randomized arm, clinically important pain level at 3 months, total GASE score, GASE score attributed to the study drug, age, HADS-anxiety, and HADS-depression. For each imputed dataset, the Mantel–Haenszel method stratified by center will be applied to estimate the RD between treatment groups. The estimates from all imputed datasets are then combined using Rubin’s rules, which account for both within-imputation variance (variability within each dataset) and between-imputation variance (variability across datasets due to imputation). This produces a pooled RD estimate, a 95% confidence interval, and a p-value that incorporate the uncertainty from missing data.

### 6.2.5 Sensitivity analyses

#### **ITT complete-case analysis**

The same model as in the primary analysis will be conducted but using a complete-case analysis approach (no imputation).

#### **Per-protocol set**

The same model as in the primary analysis will be conducted but following a per protocol principle.

## 6.3 Secondary endpoints

### 6.3.1 Generic Assessment of Side Effects

The Generic Assessment of Side Effects (GASE) will be assessed 10 days post-surgery in the full analysis set, imputing missing data with the PMM method, evaluating:

- The total GASE score (overall side effect burden)
- The GASE score attributed to the study drug

Both calculations of the GASE score will be described in each study arm, including the frequency count and percentage of patients with null GASE score.

For each calculation of the GASE score, the effect of treatment will be tested using Compound Poisson–Gamma (Tweedie) regression model, which accounts for the semi-continuous nature of the data, characterized by excess zeros and a positively skewed continuous component. The model will include fixed effects for treatment (Pregabalin vs. Placebo) and the center. Comparisons between groups will be expressed as mean difference on the response scale, with corresponding two-sided 95% confidence intervals and *p*-values.

### 6.3.2 Other secondary outcome measures

The following secondary outcomes will be reported using only descriptive statistics.

#### **1. Study Drug**

- Evening before surgery study drug dose
- Morning of surgery study drug dose
- Evening of surgery study drug dose
- Morning after surgery study drug dose
- Are you still taking the study medication? (D10)
- Did you discontinue because of side effects? (D10)
- Were you able to take all tablets as planned? (D13)
- How many tablets remain in the blister? (D13)
- Did you discontinue because of side effects? (D13)
- Drug kit retrieved from patient (D20)

#### **2. Postoperative Analgesia / Medications**

- paracetamol\_d1 – Paracetamol (yes/no)
- ibuprofen\_d1 – Ibuprofen (yes/no)



- metamizol\_d1 – Metamizol (yes/no)
- otehrnaids\_d1 – Other NSAIDs (yes/no)
- tramadol\_d1 – Tramadol (yes/no)
- morphine\_d1 – Morphine (yes/no)
- other\_opioids\_d1 – Other opioids (specify)

### 3. New Symptom / Side Effect Occurrence Questionnaire (D1)

- Are you currently experiencing any new symptoms?
- Do you attribute this symptom?
- How unpleasant are these new symptoms overall?

### 4. Blinding / Subjective Treatment Perception

- Do you have side effects which you attribute to the study drug?
- On a scale of 0-10, how unpleasant are those side effects altogether?
- Do you think you received real drug or placebo?

### 5. Pain Diary Results

- Pain diary retrieved from patient
- Mean pain at rest on day 6
- Mean pain when moving on day 6
- Did patient take pain killer on day 6?
- Mean pain at rest on day 10
- Mean pain when moving on day 10
- Did patient take pain killer on day 10?

### 6. Hospitalisation

- Length of hospitalisation (Discharge date – Day of operation)

### 7. Pain intensity

- What was your mean pain intensity in the surgical area during the last week, on a 0-10 scale? (D30)
- What was the mean pain intensity during the last week of your pain at the site of surgery (breast, axilla, arm) at rest? (M3, M6, M12)
- What was the mean pain intensity during the last week of your pain at the site of surgery (breast, axilla, arm) when moving your arm? (M3, M6, M12)
- Do you consider your pain important (i.e., interfering with your life)? (M3, M6, M12)
- In retrospect, do you consider the preventive treatment worthwhile, given your pain and your experience of taking the medication? (M3)
- Are you taking analgesics for your pain? (M3, M6, M12)
- Acute pain intensity and patient-reported pain outcome at 24h from the PAIN-Out questionnaire
- Pain intensity at rest and movement, pain interference from the Brief Pain Inventory (BPI) (M3, 6M and M12)
- Incidence of neuropathic pain defined as a DN4-self-evaluation score  $\geq 4$  (M3, M6 and M12)



#### 8. Postoperative treatment

- Whether a revision was needed within 30 days postoperatively
- Whether radiotherapy (RX) was performed within 12 months postoperatively
- Whether chemotherapy was performed within 12 months postoperatively
- Whether hormonotherapy was performed within 12 months postoperatively
- Whether aromatase inhibitor therapy was performed within 12 months postoperatively

#### 9. Expectations about treatment benefits

- "How much pain do you expect 3 months from now?"
- "How efficient do you think the study treatment is against the long-term pain (i.e. what is the success rate in %)?",
- "How well do you think it will work for you? (i.e. reduction in pain score 0-10)"

#### 10. Beliefs about treatment attribution

- "Do you think that you received the real drug or the placebo?"

#### 11. Completion of treatment

- Retention rates of pregabalin treatment (D10), defined as the still taking the study treatment as planned and having not stopped taking it due to side effects
- Stopping study treatment due to side effects attributed to the treatment (D10)

### 6.4 Baseline analysis

All participants who were randomized (ITT population) will be included in the baseline descriptive analysis. Baseline variables will include demographics, preoperative assessments, operative assessments, and postoperative medications.

Demographic variables comprise age, sex, body mass index (BMI), ASA status, and the center of recruitment.

Preoperative assessments include HADS (Hospital Anxiety and Depression Scale) depression and anxiety scores, PCS (Pain Catastrophizing Scale) rumination, magnification, and helplessness scores, PANAS (Positive and Negative Affect Schedule) positive and negative affect scores, LOT-R (Revised Life Orientation Test) scores, PHQ-15 (Patient Health Questionnaire) scores, and a preoperative questionnaire assessing expected pain perception, preoperative anxiety, and analgesic expectations.

Anesthesia assessments include volatile anesthetics (e.g., sevoflurane, desflurane), total intravenous anesthesia (continuous propofol), ketamine as an adjuvant (including total dose in mg if administered), wound infiltration, PECs block or similar, paravertebral block, and epidural.

Surgery / operative procedure assessments include tumorectomy or quadrantectomy, mastectomy, sentinel node excision, axillary lymph node clearance, reconstruction with breast implant, reconstruction without implant (breast correction), prophylactic surgery (BRCA2+), and intraoperative radiotherapy.

Operative (D0) and postoperative (D1) analgesia / medication assessments include paracetamol, ibuprofen, metamizol, other NSAIDs, tramadol, morphine, other opioids (specify), and other non-opioid medications (specify). These will be reported distinctively for both operative and postoperative. For each analgesic, the number of times they are prescribed will be reported. In addition, for each participant the total number of analgesics will be calculated and the frequency of each number of analgesics (including 'other' if the any text is written for other analgesic) will be reported. All the open

text written as other analgesic will be reported and the count and proportion of times it was filled in will also be reported.

## 6.5 Safety endpoints

Safety endpoints will be summarized descriptively in the safety population according to the following grouping: 1) received exclusively Pregabalin, 2) received exclusively placebo, and 3) received both Pregabalin and placebo due to a switch in treatment. The analyses will include:

- **Serious adverse events (SAEs):** Nature of event, unexpectability, and relationship to the study drug, from the first dose until the last study visit. Data will be summarized as the number and percentage of patients experiencing each SAE, as well as the total number of events.
- **Pregabalin-related side effects:** Only for patients using Pregabalin: experience of “new symptoms” attributed to the study drug at the first postoperative day (D1) and assessment of common central nervous system effects: dizziness, blurred vision, and fatigue at postoperative day 10 (including the respective items of the GASE questionnaire). Data will be summarized as the number and percentage of patients experiencing each side effect, and the total number of events.
- **Treatment discontinuation:** Number of patients who stop study treatment due to adverse events or intolerance.
- **SUSAR:** Number of SUSAR occurring during the study.
- **Deaths:** Number of deaths occurring during the study.

## 6.6 Laboratory endpoints

Not applicable.

## 6.7 Ancillary study

The effect of information on the total GASE score per patient will be evaluated descriptively.

### Outcomes

- Retention rates at day 10 (compliance\_e\_d10)
- Risk of stopping treatment due to side effects attributed to the treatment
- “New symptom” at d1 attributed to the study drug at the first postoperative day
- GASE total score and score attributed to the study drug at d10
- GASE items specific to pregabalin (dizziness, blurred vision, fatigue) at d10

For each outcome, a description will be reported for each of the four information conditions.

## 6.8 Assessment of Statistical Assumptions

**Independence of error from predictors.** The residuals will be plotted against each predictor to visually assess the independence between the two. In case of a violation of the assumption, the trial statistician is responsible to provide an appropriate alternative.

**Non-zero variance.** Any independent variable in a regression with a zero variance will not be used in the model.

## 6.9 Interim Analyses

An interim analysis was performed when data for the primary outcome were available for 59 patients in the placebo (19/59 (32.2%) had a clinically important pain) and for 62 patients in the Pregabalin arm (13/62 (21.0%) had a clinically important pain). The risk difference was 11.2%. The stopping rule was a difference in the incidence of the primary outcome of less than 8.2% (futility) or more than 19.9% (superiority). The results led to the continuation of the trial.

## 6.10 Statistical Software

Statistical analyses will be conducted on the software R, version 4.2.2 or above.

## 7. References

- Lipsitz, S. R., Dear, K. B. G., Laird, N. M., & Molenberghs, G. (1998). Tests for homogeneity of the risk difference when data are sparse. *Biometrics*, 54, 148–160.
- Pham, T. M., Kahan, B. C., Lopes, A., Rashid, M., Hoskin, P. J., & White, I. R. (2025). Estimating treatment effects in trials with outcome data truncated by death: A case study on aligning estimators with estimands. *Clinical Trials*, 17407745251360645.